School of Electrical Engineering and Computer Sciences
National University of Sciences and Technology

DEPARTMENT OF Computing

Industry Advisory Board Meeting Minutes

Held on November 13th, 2023

In attendance

S.No.	Name	Designation/Organization/Company
1.	Dr. Ajmal Khan (In-Chair)	Principal, NUST-SEECS
2.	Dr. Rafia Mumtaz	Professor, Head SE Department, NUST-SEECS
3.	Dr. Muhammad Bilal Ali	Assistant Professor, NUST-SEECS
4.	Dr. Hashir Moheed Kiani	Assistant Professor, NUST-SEECS
5.	Dr. Daud Adullah Asif	Assistant Professor, NUST-SEECS
6.	Mr. Taufique ur Rehman	RVF, NUST-SEECS
7.	Ms. Zunera Zahid	Lecturer, NUST-SEECS
8.	Ms. Mehvish Kiran	Lab Engineer, NUST-SEECS
9.	Mr. Umer Nasim	Software Engineer/ Department Coordinator, NUST-SEECS
10.	Mr Shamyl Bin Mansoor	CTO, LearnOBots
11.	Mr. Usman Nazir	S&P Global
12.	Mr. Imran Khan	Quixel
13.	Mr. Asim Shah	Info Affinity
14.	Mr. Saifullah	We R Play
15.	Mr. Usama Bin Ahmad	S&P Global
16.	Mr. Qaiser Rafiq	S&P Global
17.	Mr Abbas Wahab	Institute of Skill Development

Venue: SEECS Conference Room / Online

Meeting time: 1500 hrs – 1630 hrs

Agenda Points / Discussion

Dr. Ajmal Khan briefed the house about the department, the degree programs, the vision and mission of the department and how the curriculum is designed to achieve this vision and mission.

- Dr Rafia shared the agenda points of the meeting.
- Dr Hashir presented the agenda points (attached in Annex-B).
- Dr Hashir presented the PEC observation, highlighting correcting the PLO to PEO mapping. House endorsed the PLO to PEO mapping.
- Dr Hashir presented the PEC observation in which PEC highlighted that internship feedback forms should be updated. He presented the updated form. IAB members endorsed the updated form. The updated form is attached as Annex-A.
- Dr Bilal discussed the CEP related to industrial use cases. He asked the house to share specific industrial complex engineering problems that FMs can assign to the students in their courses. House decided that a platform will be made where industrial participants will provide CEPs related to industry.
- Dr Hashir presented the PEC observations where they highlighted that industry liaison is very weak. HoD SE recommended that an industry liaison officer engage more industries that can offer more FYDP.
- Dr Hashir opened the house for discussion where all members could give their suggestions regarding how the industry can more actively participate in academia. House decided that more industrial talks and training by the industry would be organised in the future.
- Dr. Daud emphasized that Q5 in the alumni survey form is ineffective in evaluating PEO-2. Nevertheless, this is unrelated to the PEO-2 statement. Consequently, it has been relocated to PEO-3, where it aligns appropriately with its KPI (attached in Annex-C).
- Dr Rafia appreciated the feedback from the industry and mentioned that the department would make due efforts to implement the suggested points.

DoC Department Coordinator
November 13, 2023

DLQEC SEECS

Annex-A

School of Electrical Engineering and Computer Sciences
National University of Sciences and Technology

Student Evaluation by Internship Employer

The purpose of this assessment is to obtain employer's input on the performance of the internee for during their internship.

Student Name: _____

Internship Period From: _____To: _____

Employer/Department:_____

Name of Supervisor:_____

Please rate the student on the following characteristics. For the attributes not applicable to this assessment or for insufficient information to evaluate that item, please check N/A.

A : Excellent (5) **B : Very Good(4)** **C : Good (3)** **D : Fair (2)** **E : Poor (1)**

S No.	Attributes	A (5)	B (4)	C (3)	D (2)	E (1)	N/A
1	Level of engineering knowledge and ability to link theory to practice in solving real world problems.						
2	Ability of problem analysis in solving complex engineering problems.						
3	Design ability or development of solutions to given problems.						
4	Investigating problems by synthesizing information to arrive at meaningful conclusions.						
5	Ability to utilize modern engineering and IT tools in practice.						
6	Ability to address societal, health, safety, legal and cultural issues in a professional manner.						

7	Mindful of environmental considerations and sustainability.						
8	Appreciates and practices professional ethics.						
9	Able to work individually as well as in team environment.						
10	Able to communicate, present and write in an effective manner.						
11	Displays project management/leadership skills with appropriate use of time and resources.						
12	Shows willingness in learning the innovations and technological developments in the field.						

Date of Evaluation: _____

Signature of Evaluator:_____

Annex-B

NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

SCHOOL OF ELECTRICAL ENGINEERING &
COMPUTER SCIENCE (SEECS)

IAB MEETING
SOFTWARE ENGINEERING DEPARTMENT

13th November 2023

AGENDA

- Washington Accord – PEC Accreditation Visit
- Issues highlighted during PEC Visit
- Curriculum and Learning Process
- Improvements Since Last IAB Meeting

WASHINGTON ACCORD PEC Accreditation Visit

- Pakistan Engineering Council (PEC) became the full signatory of Washington Accord (WA) in 2017.
- In order to maintain the quality of accreditation system in each signatory jurisdiction, WA Periodic Review of full signatories is held after every five years.
- In this context, WA representatives visited NUST on 30th October 2023.
- BE Software Engineering program at School of Electrical Engineering and Computer Science (SEECS) was among the programs reviewed for accreditation.

ISSUES HIGHLIGHTED DURING PEC VISIT

- PEO 2 mapping to PLOs was not adequate
- Need for revision of Internship feedback form
- Addition of Electives relevant to Industrial needs
- Addition of industry relevant Complex Engineering Problem to courses
- Limited engagement of industry in FYDPs
- Limited industrial trainings and industry relevant short courses

PLOs to PEO Mapping

- PEO 2: Lead diverse teams and work collaboratively to develop innovative solutions.

S. No.	PLOs	PEOs		
		1	2	3
1	Engineering Knowledge	✔		
2	Problem Analysis	✔		
3	Design/Development of Solutions	✔	✔	
4	Investigation	✔		
5	Modern Tool Usage	✔		
6	The Engineer and Society			✔
7	Environment and Sustainability			✔
8	Ethics			✔
9	Individual and Teamwork		✔	
10	Communication		✔	
11	Project Management		✔	
12	Lifelong Learning	✔		

CURRICULUM CHANGES DURING LAST TWO YEARS

Inclusion / Deletion of Courses

Inclusion of Courses	Deletion of Courses
Occupational Health and Safety (1+0)	Digital Logic Design (3+1)
Computer Architecture and Logic Design (3+1)	Computer Architecture and Organization (3+1)
Cloud Computing (2+1)	Software Requirements Engineering (3+0)
Human Computer Interaction (2+1)	
Information Security (3+0)	
Design and Analysis of Algorithms (3+0)	
Fundamentals of computer programming (3+1)	Fundamentals of programming (2+1)

6

ADDITION OF ELECTIVES DURING LAST TWO YEARS

Newly Included Core Depth Elective Courses

Software Eng- Core Depth Elective Courses		
Sr.	Title	CHs
1	Big Data Analytics	2+1
2	Deep Learning	3+1
3	Agent Based Software Engineering	2+1
4	Visual Programming	2+1
5	Real Time Systems	2+1
6	Data Encryption and Security	2+1
7	Augmented and Virtual Reality	3+1

Newly Included Multi Disciplinary Engineering Elective Courses

Multi Disciplinary Engineering Electives		
Sr	Title	CHs
1	Internet of Things	3+0
2	Embedded Systems	2+1

7

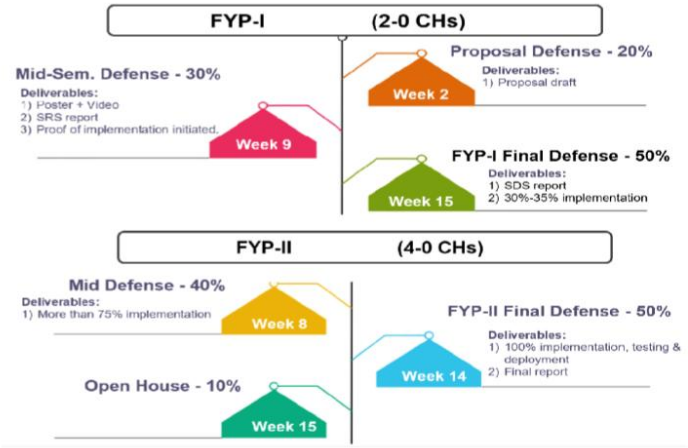
COMPLEX ENGINEERING PROBLEMS

- Complex Engineering Problems are design problems that require in-depth engineering knowledge, have conflicting requirements and have no obvious solution
- Kind of problems encountered by engineers during their professional practice
- Objective is to give them a flavor of such problems throughout their degree
- The challenge is how such industry relevant design problems can be incorporated in the curriculum

FINAL YEAR DESIGN PROJECTS

- The evaluation criteria and milestones for FYDPs have been revised
- Strict measures have been introduced at time of proposal defence to ensure quality FYPs are undertaken by students
- The evaluation rubrics of each milestone have been revised.
- Mandatory to select FYP topics that are aligned with **UN's Sustainable Development Goals**
- Industrial sponsorship of FYDPs is very limited

FYDP MILESTONES



INDUSTRIAL TRAINING

- Internship of at least 6 - 8 weeks is encouraged after completion of 2nd and 3rd year of degree.

Internship Details of the BESE Program				
Batch	**2019**	**2020**	**2021**	**2022**
BESE 2k16	76			
BESE 2k17	58	34		
BESE 2k18	38		61	
BESE 2k19			44	59
BESE 2k20			32	45
BESE 2k21				26
Total Students	**172**	**34**	**137**	**130**

11

CURRICULUM AND LEARNING PROCESS

- Periodic CQI meetings are held to improve the curriculum
- The department has revised the CLOs and BT levels of several courses
- Additional CLOs have been added covering psychomotor and affective domains to balance overall PLO mapping
- Lab rubrics have been revised to assess psychomotor and affective domains effectively
- Formal training sessions were conducted for LEs and FMs on OBE

INFRASTRUCTURE IMPROVEMENT

- The construction of a new academic block has been planned in 2023 for provisioning of about 30,000 sq. ft. space for class rooms, labs, faculty/ support offices for Dept.
- Dept. has also initiated establishment of a one general purpose lab and one additional FYDP lab to facilitate practical learning of students .
- In addition 2 x new labs in collaboration with industry have been established - Walee-NUST centre of excellence and OneScreen research lab. The industrial partners have also provided more than PKR 5M value equipment to SEECS for said labs.

WALEE-NUST CENTRE OF EXCELLENCE

- SE department has initiated collaboration with Walee Technologies.
- For this purpose, a joint lab in SEECS has been established.
- This lab offers opportunities such as paid internships, training, and scholarships to students.
- In addition, faculty is financially compensated for supervising the company projects with the help of students.

14

Collaboration with Coventry University, UK

- SE department has initiated collaboration with Coventry University, UK.
- For this purpose, a joint lab in SEECS has been established with the name of NUST-Coventry Internet of Things Lab (NCIL).
- This lab will offer opportunities such as internships at Coventry University, joint projects, joint supervision, joint curriculum development and upgradation.
- In addition, fully funded PhD opportunities will be offered as well at Coventry University, UK

15

SOFT SKILLS

- Oral presentation is now mandatory in all courses to improve communication skills of students
- Mock interviews are arranged for final year students to improve their interviewing skills

16

INDUSTRY TALKS, WEBINARS AND WORKSHOPS

- Industry Talks, Webinars and Workshops are held quite frequently to keep the students apprised about latest industrial trends

17

DEVOPS CULTURE

- The students are encouraged to use software version control tools, automated build tools and IDEs in their projects and assignments
- The students are offered a course on Open Source Systems

18

PARTICIPATION IN COMPETITIONS

- Students are encouraged to participate in national and international competitions like Microsoft Imagine Cup, Huawei ICT Innovation Competition, Google's coding competitions, Cisco Global Problem Solver Challenge etc

- Recently SE department FYDPs have won first position and a merit award in the categories of IoT & Digital Systems, and Software Applications at 20th Edition of COMPEC 2023.

19

IMPROVEMENTS SINCE LAST IAB MEETING (AUG, 2023)

- **Industry suggested inclusion of course on Virtual and Augmented Reality**
  - **Action**: The course has been included as an elective and offered to final year students in Fall 23.
- **Industry suggested more focus on core programming skills**
  - **Action**: Fundamentals of Programming course has been changed from 3 CHs to 4 CHs
- **Industry suggested that product-based projects should be encouraged.**
  - **Action**: Students are encouraged to take projects in collaboration with industry. More efforts need to be put in this regard as industry collaboration of FYDPs is still very limited.
- **Industry suggested that FYDPs should be encouraged to file patents**
  - **Action**: Students are encouraged to file patents for their FYPs. NUST IP office provides full support in this regard.

20

Input Required from Industry

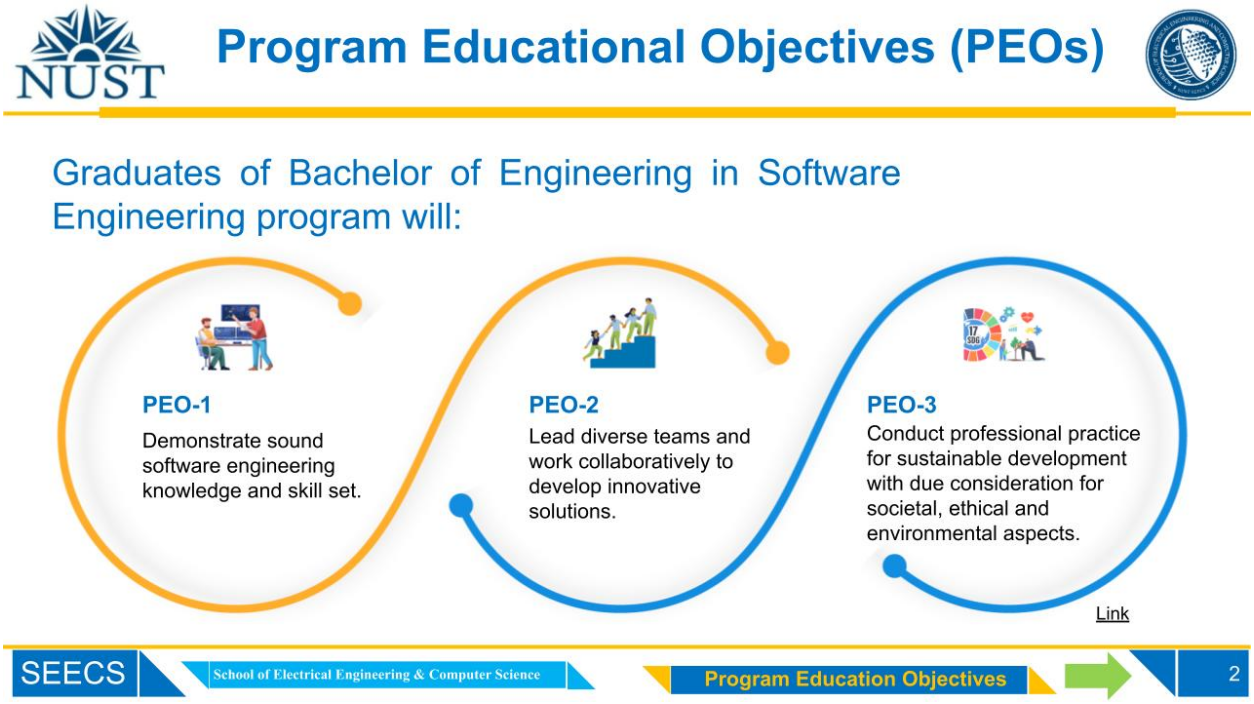
- Suggest topics / content for addition in curriculum to align it with industry needs
- Deliver guest lectures for selected courses
- Offer internships and trainings for students and short-term placements for faculty
- Engage students in Industry projects by offering FYDPs
- Give advice on incorporation of Complex Engineering Problems
- Give advice on improvement of lab component of courses

21

THANK YOU

Annex-C

Revised PEOs Attainment KPIs



PEO Assessment and KPIs

Assessment	Measurement tool	Method	KPI	When measured
PEO-1	Employer Survey	Indirect	Feedback (Q 1 & 2) ≥ 60%	4 years after graduation
	Alumni Survey		Feedback (Q 1 & 2) ≥ 60% Feedback (Q 6 & 7) ≥ 50%	
PEO-2	Employer Survey		Feedback (Q 3, 4 & 5) ≥ 60%	
	Alumni Survey		Feedback (Q 3 & 4) ≥ 60%	
PEO-3	Employer Survey		Feedback (Q 6 & 7) ≥ 60%	
	Alumni Survey		Feedback (Q5, 8 & 9) ≥ 50%	

PEO ATTAINMENT KPIs

PEO # 1: Demonstrate sound software engineering knowledge and skill set.	
KPI	**Measurement Tool**
60% of respondent employers are satisfied with the technical knowledge of employed graduates	Employer Survey (Q1)
60% of respondent employers are satisfied with the competence and skills of employed graduates	Employer Survey (Q2)
60% of respondent graduates give positive feedback on their knowledge compared to their peers	Alumni Survey (Q1)
60% of respondent graduates give positive feedback on their competence compared to their peers	Alumni Survey (Q2)
50% of respondent graduates are employed within one year of their graduation	Alumni Survey (Q6)
50% of respondent graduates are pursuing higher education	Alumni Survey (Q7)

PEO ATTAINMENT KPIs

PEO #2: Lead diverse teams and work collaboratively to develop innovative solutions.	
KPI	**Measurement Tool**
60% of respondent employers are satisfied with the communication skills of employed graduates	Employer Survey (Q3)
60% of respondent employers are satisfied with the teamwork skills of employed graduates.	Employer Survey (Q4)
60% of respondent employers are satisfied with the independent project-handling skills of employed graduates	Employer Survey (Q5)
60% of respondent graduates are involved in research and/or innovation activities	Alumni Survey (Q3)
60% of respondent graduates have handled projects independently	Alumni Survey (Q4)

PEO ATTAINMENT KPIs

PEO #3: Conduct professional practice for sustainable development with due consideration for societal, ethical, and environmental aspects.	
KPI	**Measurement Tool**
60 % of respondent employers are satisfied with adherence to ethical values by employed graduates.	Employer Survey (Q6)
60 % of respondent employers are satisfied with the discharge of professional obligations of employed graduates.	Employer Survey (Q7)
50% of respondent graduates have incorporated UN SDGs in their projects/services / solutions/products etc.	Alumni Survey (Q8)
50% of respondent graduates are involved in professional development activities	Alumni Survey (Q5)
50% of respondent graduates are involved in projects addressing societal and/or environmental issues	Alumni Survey (Q9)

SEECS School of Electrical Engineering & Computer Science